58. LIBRARY BORROWING AND FINE RECONCILIATION

1. AIM

To compute overdue fines for returned library books, classify return status, and calculate total fine collection.

2. PROBLEM STATEMENT

A campus library imposes a fine of Rs. 2 per late day on overdue books. Write a Python program that accepts book IDs and overdue days, calculates individual fines, checks overdue status, and prints aggregate fines.

3. OBJECTIVES

- 1. Input number of returned books.
- 2. Calculate fine = late_days * 2.
- 3. Flag overdue books vs on-time returns.
- 4. Sum total fines accumulated.

4. ASSUMPTIONS & REQUIREMENTS

Assumptions:

- Late days is non-negative.
- Fine rate is constant at Rs. 2/day.

Requirements:

- Python 3.x Interpreter
- Loop accumulator pattern

5. ALGORITHM

Step 1: Read number of books n. Initialize total_fine = 0.

Step 2: Loop n times:

a. Read book ID and late days.

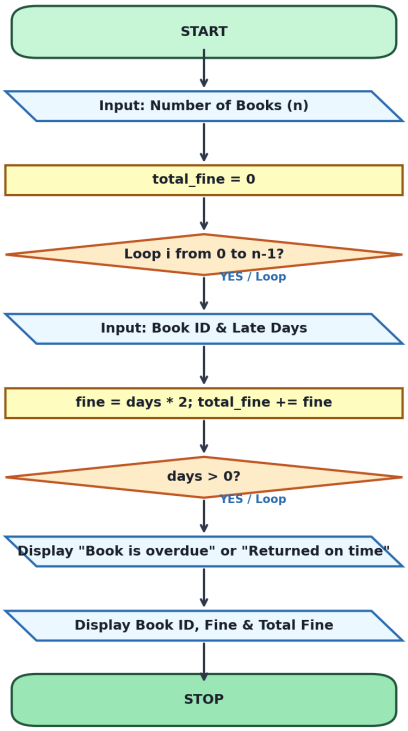
b. fine = days * 2; total_fine += fine.

c. Print book ID and fine.

d. If days > 0: print 'Book is overdue', else print 'Book returned on time'.

Step 3: Print total_fine. Stop.

6. PROGRAM FLOWCHART



7. PYTHON PROGRAM (library.py)

```
n = int(input("Enter number of books: "))
total_fine = 0

for i in range(n):
    book = input("Enter book ID: ")
    days = int(input("Enter late days: "))

    fine = days * 2
    total_fine += fine

    print("Book ID:", book)
    print("Fine: Rs.", fine)

    if days > 0:
        print("Book is overdue")
    else:
        print("Book returned on time")

print("Total fine: Rs.", total_fine)
```

8. EXPLANATION OF THE CODE

- **fine = days * 2:** Calculates individual fine based on fixed rate.
- **total_fine += fine:** Accumulates total library fine collection.
- **if days > 0:** Evaluates whether book exceeded loan period.

9. TRACING TABLE

Book ID	Late Days	Individual Fine	Total Fine Accumulator	Status
123	2	2 * 2 = Rs. 4	Rs. 4	Book is overdue

10. TEST CASES

Test Case	Input	Expected Result	Actual Result
TC 1 (Overdue)	1 book: ID 123, 2 late days	Fine: Rs. 4 Book is overdue Total fine: Rs. 4	Fine: Rs. 4 Book is overdue Total fine: Rs. 4
TC 2 (On Time)	1 book: ID 456, 0 late days	Fine: Rs. 0 Book returned on time Total fine: Rs. 0	Fine: Rs. 0 Book returned on time Total fine: Rs. 0
TC 3 (Multiple Books)	2 books: ID A (3 days), ID B (0 days)	Book A Fine: 6 Book B Fine: 0 Total fine: Rs. 6	Book A Fine: 6 Book B Fine: 0 Total fine: Rs. 6

11. EXPECTED OUTPUT

```
Enter number of books: 3
Enter book ID: 123
Enter late days: 2
Book ID: 123
Fine: Rs. 4
Book is overdue
Enter book ID:
```

12. ACTUAL OUTPUT

```
Enter number of books: 3
Enter book ID: 123
Enter late days: 2
Book ID: 123
Fine: Rs. 4
Book is overdue
Enter book ID:
```

13. PROGRAM EXECUTION SCREENSHOT

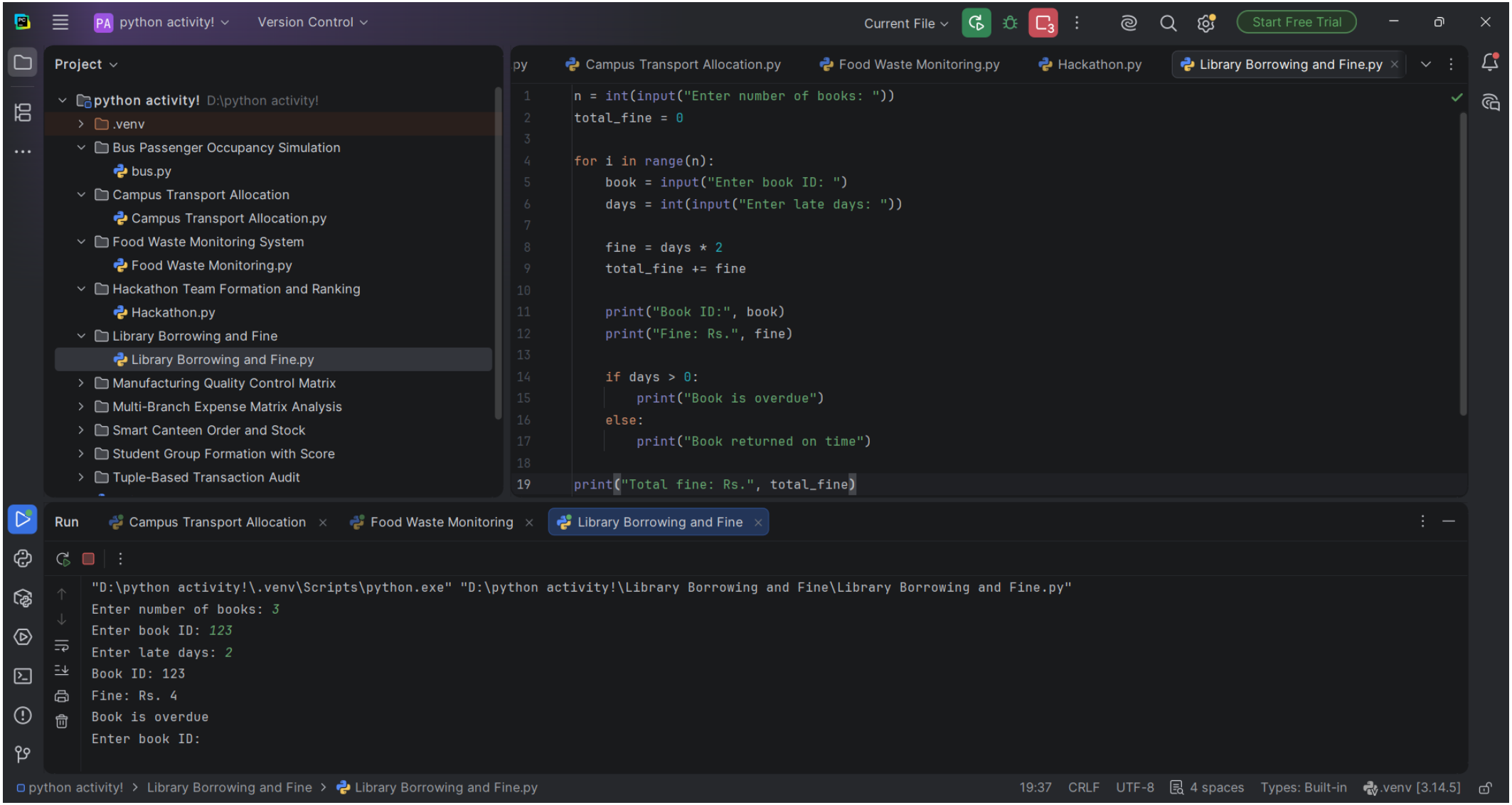
[Screenshot to Be Added]

14. RESULT / CONCLUSION

The Python program library.py was executed successfully. Overdue fine amounts and total reconciliation were correct.

58 Library Fine Reconciliation

Actual Input and Output – PyCharm Run Screenshot



Screenshot includes the program code and the console inputs/outputs.